MATH 152
Mrs. Bonny Tighe

EXAM IIIA

PART TWO
11.8-11.11, 10.5
50 points

NAME Solutions

TA_____ Wed.12/1/2021

1. (5 points) a) Find a Taylor polynomial of degree 3 for $f(x) = \dfrac{1}{x\sqrt{x}}$ centered at $a = 4$

$f(x) = x^{-3/2}$ $\quad f(4) = 4^{-3/2} = \frac{1}{8}$

$f'(x) = -\frac{3}{2}x^{-5/2}$ $\quad f'(4) = -\frac{3}{2}\cdot\frac{1}{32} = \frac{-3}{64}$

$f''(x) = \frac{15}{4}x^{-7/2}$ $\quad f''(4) = \frac{15}{4}\cdot\frac{1}{2^7}$

$f'''(x) = \frac{-105}{8}x^{-9/2}$ $\quad f'''(4) = \frac{-105}{8}\cdot\frac{1}{2^9}$

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(4)(x-4)^n}{n!}$$

$$T_3(x) = \frac{1}{8} - \frac{3}{64}(x-4) + \frac{15}{4(2^7)2!}(x-4)^2 - \frac{105}{8(2^9)3!}(x-4)^3$$

b) (4 points) Use it to approximate $\sqrt{4.1}$.

$4.1 - 4 = .10$

$$\sqrt{4.1} \approx \frac{1}{8} - \frac{3}{64}(.1) + \frac{15}{4(2^7)2!}(.1)^2 - \frac{105}{8(2^9)3!}(.1)^3$$

$8.5 < x < 9.5$

$-0.5 < x-9 < 0.5$

$\frac{105}{9}$

945

c) (4 points) How accurate is this approximation when $|x-9| < 0.5$

If $|f^{(n+1)}(x)| \le M$ for $|x-a| \le d$, then the remainder $R_n(x) \le \dfrac{M}{(n+1)!}|x-a|^{n+1}$ for $|x-a| \le d$

$d = .5$

$f''''(x) = \frac{945}{16}x^{-11/2}$

Largest when $x = 8.5$

$M = \dfrac{945}{16(8.5)^{11/2}}$

$$R_3(x) \le \frac{945}{16(8.5)^{11/2}}\frac{.51^4}{4!}$$

$3\cdot5\cdot7\cdot9$

15

63

945

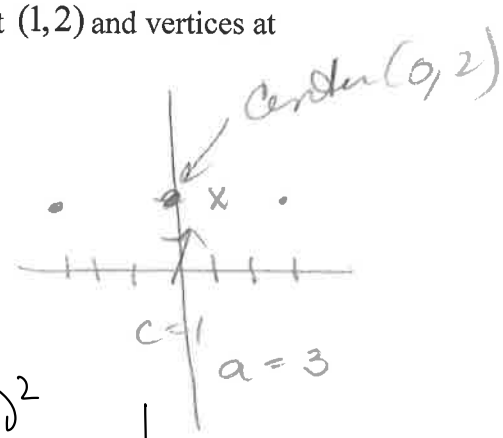
90

945

4

2. (4 points) Find the equation of the ellipse with and focus at $(1,2)$ and vertices at $(3,2)$ and $(-3,2)$

$$\boxed{\frac{x^2}{9}+\frac{(y-2)^2}{8}=1}$$



$c^2=a^2+b^2$
$1=9-b^2$
$b^2=8$

$$\frac{x^2}{9}+\frac{(y-2)^2}{b^2}=1$$
$=8$

$c^2=a^2+b^2$
$1=9+b^2$

3. (5 points) Find the Interval and Radius of convergence for the power series

$$\sum_{n=0}^{\infty}\frac{(-1)^n(x-2)^n}{\sqrt{n+3}}$$

Ratio Test

$$\lim_{n\to\infty}\left|\frac{(x-2)(x-2)^n}{\sqrt{n+4}}\cdot\frac{\sqrt{n+3}}{(x-2)^n}\right| = \lim_{n\to\infty}\left|(x-2)\frac{\sqrt{n+3}}{\sqrt{n+4}}\right| = |x-2|<1$$

$-1<x-2<1$
$1<x<3$

$R=1 \quad I=(1,3]$

Test end points

$x=1 \quad \sum_{n=0}^{\infty}\frac{(-1)^n(-1)^n}{\sqrt{n+3}} = \sum_{n=0}^{\infty}\frac{1}{\sqrt{n+3}}$ Diverges by Limit Comparison Test to p-series $p=1/2\le 1$

$x=3 \quad \sum_{n=0}^{\infty}\frac{(-1)^n(1)^n}{\sqrt{n+3}} = \sum_{n=0}^{\infty}\frac{(-1)^n}{\sqrt{n+3}}$ conditionally convergent by AST

4

4. (4 points) Find the vertex, focus, and directrix. Sketch the graph

$y^2 + 4y + 4x + 8 = 0$

$(y+2)^2 = -4x - 8 + 4$

$(y+2)^2 = -4x - 4 = -4(x+1)$

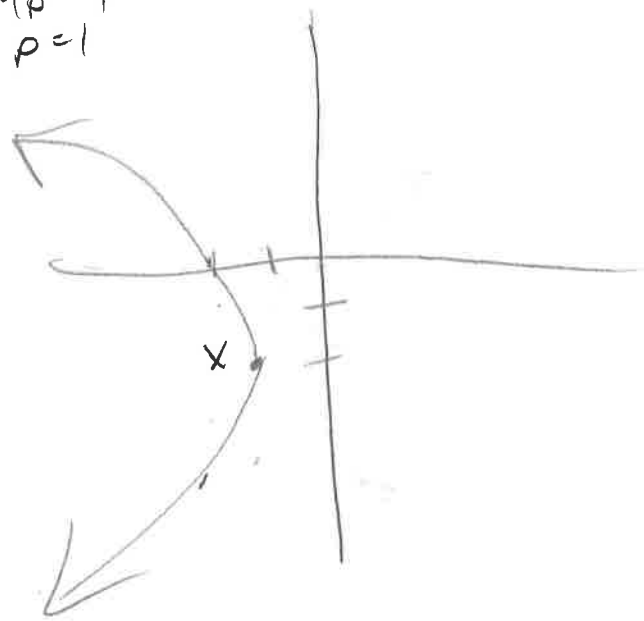
center $(-1,-2)$

focus $(-2,-2)$

directrix $x = 0$

$4p = 4$

$p = 1$



5

5. (4 points) Find the vertices, foci, asymptotes, and center. Sketch the graph.

$4(y-2)^2 - 9(x-1)^2 = 36$

$\frac{(y-2)^2}{9} - \frac{(x-1)^2}{4} = 1$

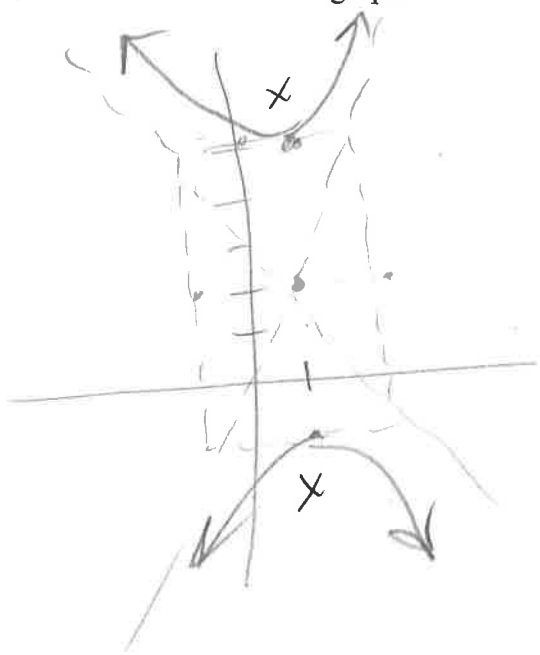
center $(1,2)$

foci $(1, 2 \pm \sqrt{13})$

vertices $(1, 2 \pm 3)$

asymptotes

$(y-2) = \pm 3/2 (x-1)$



$c^2 = a^2 + b^2$

$c^2 = 9 + 4$

$13 = c^2$

5. (7 points) Use the Binomial Theorem to find the function as a power series.

$$f(x)=\frac{1}{\sqrt[3]{1-4x^2}} = \left(1+(-4x^2)\right)^{-1/3} \qquad k=-1/3 \qquad \sum_{n=0}^{\infty} \square \frac{(-4x^2)^n}{n!}$$

$$\square = 1 + k + k(k-1) + k(k-1)(k-2)$$

$$= 1 - 1/3 - 1/3(-4/3) - 1/3(-4/3)(-7/3) - 1/3(-4/3)(-7/3)(-10/3)$$

$$\underset{n=0}{1} - \underset{n=1}{\frac{1}{3}} + \underset{n=2}{\frac{1\cdot 4}{3^2}} - \underset{n=3}{\frac{1\cdot 4\cdot 7}{3^3}} + \underset{n=4}{\frac{1\cdot 4\cdot 7\cdot 10}{3^4}} \qquad \frac{(-1)^n(1\cdot 4\cdot 7\cdots(3n-2))}{3^n}$$

Can't be negative

$$1 + \sum_{n=1}^{\infty} \frac{(-1)^n(1\cdot 4\cdot 7\cdots(3n-2))}{3^n} \frac{(-1)^n 4^n x^{2n}}{n!}$$

$$\boxed{1 + \sum_{n=1}^{\infty} \frac{(1\cdot 4\cdot 7\cdots(3n-2))\,4^n}{3^n\, n!} x^{2n}}$$

6. (5 points) Evaluate the integral as a power series and approximate to three non zero terms. $\int_0^{0.2} \frac{x^2}{4+x^2}\,dx$

$$\int_0^{0.2} \frac{\frac{x^2}{4}}{1-\left(\frac{-x^2}{4}\right)}\,dx = \int_0^{0.2} \sum_{n=0}^{\infty} \frac{x^2}{4}\left(\frac{-x^2}{4}\right)^n dx$$

$$\int_0^{0.2} \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+2}}{4^{n+1}}\,dx = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+3}}{(2n+3)4^{n+1}} \Bigg|_0^{0.2}$$

$$\frac{x^3}{3(4)} - \frac{x^5}{5(4^2)} + \frac{x^7}{7(4^3)} \Bigg|_0^{0.2}$$

$$\frac{(.2)^3}{12} - \frac{(.2)^5}{5(16)} + \frac{(.2)^7}{7(64)} - 0$$

7. (10 points) Find the Taylor series for $f(x) = \dfrac{1}{(x-1)^3}$ centered at

a = 3

$f(x) = (x-1)^{-3}$ $\quad f(3) = 2^{-3} = \frac{1}{8}$

$f'(x) = -3(x-1)^{-4}$ $\quad f'(3) = \frac{-3}{2^4}$

$f''(x) = 4 \cdot 3(x-1)^{-5}$ $\quad f''(3) = \frac{4 \cdot 3}{2^5}$

$f'''(x) = -5 \cdot 4 \cdot 3(x-1)^{-6}$ $\quad f'''(3) = \frac{-5 \cdot 4 \cdot 3}{2^6}$

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(3)(x-3)^n}{n!}$$

$$\frac{1}{8} - \frac{3}{2^4} + \frac{4 \cdot 3}{2^5} - \frac{5 \cdot 4 \cdot 3}{2^6}$$

$$\underset{n=0}{\frac{2!}{2^4}} - \underset{n=1}{\frac{3!}{2^5}} + \underset{n=2}{\frac{4!}{2^6}} - \underset{n=3}{\frac{5!}{2^7}}$$

$$\frac{(-1)^n (n+2)!}{2^{n+4}}$$

$$\sum_{n=0}^{\infty} \frac{(-1)^n (n+2)(n+1)\cancel{(n!)}(x-3)^n}{2^{n+4}\,\cancel{n!}}$$

$$\boxed{\sum_{n=0}^{\infty} \frac{(-1)^n (n+2)(n+1)}{2^{n+4}} (x-3)^n}$$

8. (5 points) Use series to evaluate the limit $\lim_{x\to 0}\dfrac{x^2-\arctan(x^2)}{e^{x^3}-1-x^3}$

$$\tan^{-1}(x^2)=\sum_{n=0}^{\infty}\frac{(-1)^n(x^2)^{2n+1}}{2n+1}=\sum_{n=0}^{\infty}\frac{(-1)^n x^{4n+2}}{2n+1}$$

$$e^{x^3}=\sum_{n=0}^{\infty}\frac{(x^3)^n}{n!}=\sum_{n=0}^{\infty}\frac{x^{3n}}{n!}$$

$$\lim_{x\to 0}\frac{x^2-\left[x^2-\frac{x^6}{3}+\frac{x^{10}}{5}-\right]}{1+\frac{x^3}{1}+\frac{x^6}{2!}+\frac{x^9}{3!}-1-x^3}$$

$$\lim_{x\to 0}\frac{\left[\frac{x^6}{3}-\frac{x^{10}}{5}+\frac{x^{14}}{7},\dots\right]}{\left[\frac{x^6}{2}+\frac{x^9}{3!}+\frac{x^{12}}{4!}\dots\right]}=$$

$$\lim_{x\to 0}\frac{\cancel{x^6}\left[\frac{1}{3}-\frac{x^4}{5}-\frac{x^9}{7}\dots\right]}{\cancel{x^6}\left[\frac{1}{2}-\frac{x^3}{3!}-\frac{x^6}{4!}\dots\right]}=$$

$$\lim_{x\to 0}\frac{\frac{1}{3}}{\frac{1}{2}}=\frac{2}{3}$$

MATH 152
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EXAM III

PART TWO
11.8-11.1,10.5
50 points

NAME Solutions

TA_____ Wed.12/1/2021

1. (~~5~~ 4 points) Find the vertex, focus, and directrix. Sketch the graph
$x^2+4x+2y+6=0$

$x^2+4x+4 = -2y-6+4$

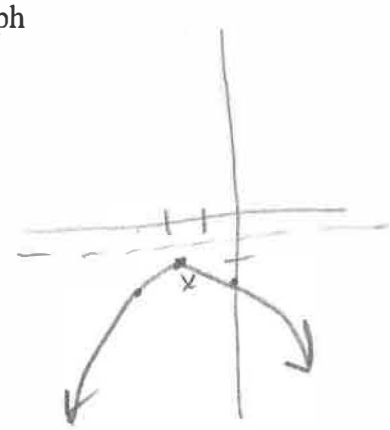
$(x+2)^2 = -2y-2$

$(x+2)^2 = -2(y+1)$

$4p = -2$

$p = 1/2$

vertex $(-2,-1)$
focus $(-2, -3/2)$
directrix $y = -1/2$



2. (~~5~~ 4 points) Use the Binomial Theorem to find the function as a power series.

$f(x) = \dfrac{1}{\sqrt[3]{1-4x^2}} = \left(1+(-4x^2)\right)^{-1/3}$

$\sum_{n=0}^{\infty} \square \dfrac{(-4x^2)^n}{n!}$

$\square = 1 + k + k(k-1) + k(k-1)(k-2)\cdots$

$= 1 - 1/3 - 1/3(-4/3) - 1/3(-4/3)(-7/3)\cdots$

$1 - \frac{1}{3} + \frac{1\cdot 4}{3^2} - \frac{1\cdot 4\cdot 7}{3^3} + \cdots \quad \frac{(-1)^n(1\cdot 4\cdot 7\cdots(3n-2))}{3^n}$

$n=0 \quad n=1 \quad n=2 \quad n=3$

$1 + \sum_{n=1}^{\infty} \dfrac{(-1)^n(1\cdot 4\cdot 7\cdots(3n-2))}{3^n} \dfrac{(-1)^n 4^n x^{2n}}{n!} =$

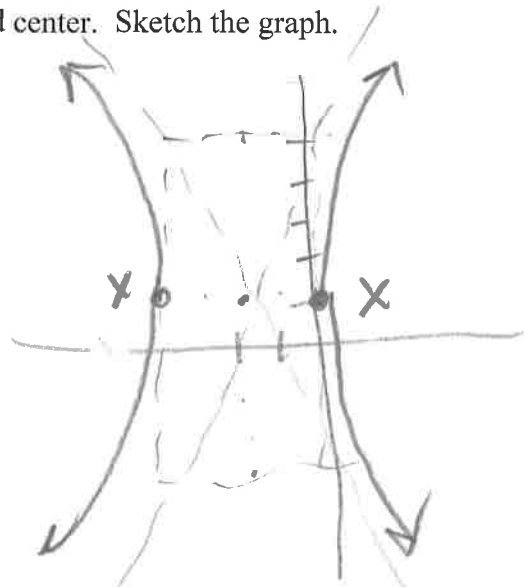
$$1 + \sum_{n=1}^{\infty} \frac{(1\cdot 4\cdot 7\cdots(3n-2))\, 4^n x^{2n}}{3^n\, n!}$$

3. (5 points) Find the vertices, foci, asymptotes, and center. Sketch the graph.

$9(x+2)^2 - 4(y-1)^2 = 36$

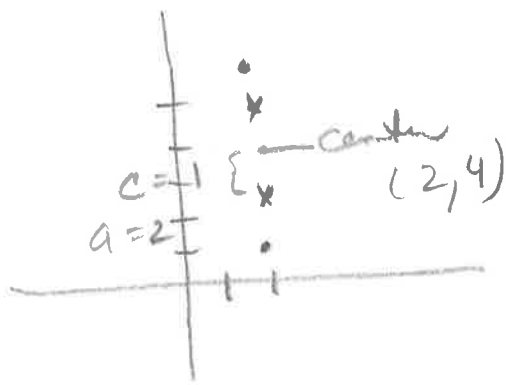
$\frac{(x+2)^2}{4} - \frac{(y-1)^2}{9} = 1$

Center $(-2, 1)$
vertices $(-2 \pm 2, 1)$
foci $(-2 \pm \sqrt{13}, 1)$
asymptotes
$(y-1) = \pm \frac{3}{2}(x+2)$



4. (4 points) Find the equation of the ellipse with foci at (2, 3) and vertex at (2, 6).

$\frac{(y-4)^2}{4} + \frac{(x-2)^2}{3} = 1$



$c^2 = a^2 - b^2$
$1 = 4 - b^2$
$b^2 = 3$

5. (5 points) Find the Interval and Radius of convergence for the power series

$\sum_{n=0}^{\infty} \frac{(-1)^n (x-2)^n}{3^n (n^2+2)}$

$\lim_{n \to \infty} \left| \frac{(x-2)(x-2)^n}{3 \cdot 3^n ((n+1)^2+2)} \cdot \frac{3^n (n^2+2)}{(x-2)^n} \right| = \lim_{n \to \infty} \left| \frac{x-2}{3} \cdot \frac{n^2+2}{(n+1)^2+2} \right| =$

$\left| \frac{x-2}{3} \right| < 1$ $\quad |x-2| < 3$

$-3 < x-2 < 3$

$-1 < x < 5$

$R = 3 \quad I = [-1, 5]$

Test end points

$x=-1$ $\sum_{n=0}^{\infty} \frac{(-1)^n (-3)^n}{3^n (n^2+2)} = \sum_{n=0}^{\infty} \frac{(-1)^n (-1)^n 3^n}{3^n (n^2+2)} = \sum_{n=0}^{\infty} \frac{1}{n^2+2}$ Convergent by comparison to a p-series

$x=5$ $\sum_{n=0}^{\infty} \frac{(-1)^n 3^n}{3^n (n^2+2)} = \sum_{n=0}^{\infty} \frac{(-1)^n}{n^2+2}$ Convergent by comparison to a p-series

x^{4n}

6. (5 points) Use series to evaluate the limit $\lim_{x\to 0}\frac{1-\cos(x^2)}{1+x^2-e^{x^2}}$

$$\cos(x^2)=\sum_{n=0}^{\infty}\frac{(-1)^n(x^2)^{2n}}{(2n)!}=1-\frac{x^4}{2!}+\frac{x^8}{4!}-\frac{x^{12}}{6!}\cdots$$

$$e^{x^2}=\sum_{n=0}^{\infty}\frac{x^{2n}}{n!}=1+x^2+\frac{x^4}{2!}+\frac{x^6}{3!}+\frac{x^8}{4!}\cdots$$

$$\lim_{x\to 0}\frac{1-\left[1-\frac{x^4}{2}+\frac{x^8}{4!}-\frac{x^{12}}{6!}\cdots\right]}{1+x^2-\left[1+x^2+\frac{x^4}{2}+\frac{x^6}{6}+\frac{x^8}{24}\cdots\right]}=$$

$$\lim_{x\to 0}\frac{\frac{x^4}{2}-\frac{x^8}{4!}+\frac{x^{12}}{6!}}{-\frac{x^4}{2}-\frac{x^6}{6}-\frac{x^8}{24}\cdots}=-1$$

7. (5 points) Find the Taylor series for $f(x)=\frac{1}{(2+x)^3}$ centered at a = 2

$$\sum_{n=0}^{\infty}\frac{f^{(n)}(2)(x-2)^n}{n!}$$

$f(x)=(2+x)^{-3}$ $\quad f(2)=\frac{1}{4^3}$

$f'(x)=-3(2+x)^{-4}$ $\quad f'(2)=\frac{-3}{4^4}$

$f''(x)=4\cdot 3(2+x)^{-5}$ $\quad f''(2)=\frac{4\cdot 3}{4^5}$

$f'''(x)=-\frac{5!}{2}(2+x)^{-6}$ $\quad f'''(2)=\frac{-5!}{2(4^6)}$

$$\underset{n=0}{\frac{1}{4^3}}-\underset{n=1}{\frac{3!}{2(4^4)}}+\underset{n=2}{\frac{4!}{2(4^5)}}-\underset{n=3}{\frac{5!}{2(4^6)}}$$

$$\frac{(-1)^n(n+2)!}{2(4^{n+3})}$$

$$\sum_{n=0}^{\infty}\frac{(-1)^n(n+2)(n+1)\cancel{n!}}{2(4^{n+3})\cancel{n!}}(x-2)^n$$

$$\sum_{n=0}^{\infty}\frac{(-1)^n(n+2)(n+1)}{2(4^{n+3})}(x-2)^n$$

8. (5 points) a) Find a Taylor polynomial of degree 3 for $f(x) = \sqrt{x}$ centered at $a = 4$.

$f(x) = x^{1/2}$ $\quad f(4) = 2$

$f'(x) = \frac{1}{2}x^{-1/2}$ $\quad f'(4) = \frac{1}{2}\left(\frac{1}{2}\right)$

$f''(x) = -\frac{1}{4}x^{-3/2}$ $\quad f''(4) = -\frac{1}{4}\cdot\frac{1}{8}$

$f'''(x) = \frac{3}{8}x^{-7/2}$ $\quad f'''(4) = \frac{3}{8}\cdot\frac{1}{32}$

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(4)(x-4)^n}{n!}$$

$$T_3(x) = 2 + \frac{1}{4}(x-4) - \frac{1}{32}\frac{(x-4)^2}{2!} + \frac{3}{256}\frac{(x-4)^3}{3!}$$

b) (4 points) Use it to approximate $\sqrt{3.9}$.

$(3.9-4) = (-.1)$

$$\sqrt{3.9} = 2 + \frac{1}{4}(-.1) - \frac{1}{32}\frac{(-.1)^2}{(2!)} + \frac{3}{256(3!)}(-.1)^3$$

c) (5 points) If $\left|f^{(n+1)}(x)\right| \le M$ for $|x-a| \le d$, then the remainder

$$R_n(x) \le \frac{M}{(n+1)!}|x-a|^{n+1} \quad \text{for } |x-a| \le d$$

How accurate is this approximation when $3.5 < x < 4.5$

$-.5 < x-4 < .5$

$|x-4| < .5$

$d = .5$

$f''''(x) = \frac{-15}{16} x^{-7/2}$

$f''''(4) = \left| \frac{-15}{16(3.5)^{7/2}} \right| \le M$

$$R_3(x) \le \frac{15}{16(3.5)^{7/2}} \frac{(.5)^4}{4!}$$

9. (5 points) Evaluate the integral as a power series to three non-zero terms.

$$\int_0^{0.5} \frac{x}{5-x^3}\,dx$$

$$\int_0^{0.5} \frac{\frac{x}{5}}{1-\frac{x^3}{5}}\,dx = \int_0^{0.5} \sum_{n=0}^{\infty} \frac{x}{5}\left(\frac{x^3}{5}\right)^n dx$$

$$\int_0^{0.5} \sum_{n=0}^{\infty} \frac{x^{3n+1}}{5^{n+1}}\,dx =$$

$$\sum_{n=0}^{\infty} \frac{x^{3n+2}}{5^{n+1}(3n+2)}\Bigg]_0^{0.5} = \frac{x^2}{5(2)} + \frac{x^5}{5^2(5)} + \frac{x^8}{5^3(8)}\Bigg]_0^{0.5}$$

$$\frac{(0.5)^2}{10} + \frac{(0.5)^5}{125} + \frac{(0.5)^8}{5^3(8)}$$

MATH 152
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EXAM III

PART ONE
11.8-11.11,10.5
50 points

NAME Solutions

TA_____ Wed. 11/25/2019

1. (3 points each) **Circle the letter for the correct answer**

a) The sum of the series $\sum_{n=0}^{\infty} \frac{(-1)^n (\pi)^{2n}}{9^n (2n)!}$

$\sum_{n=0}^{\infty} \frac{(-1)^n (\pi/3)^{2n}}{(2n)!} =$

$\cos \pi/3 = 1/2$

A) $\pi/3$

B) $3\sqrt{3}/2$

C) $\sqrt{3}/2$

D) $1/2$

E) Some other answer

b) The radius of convergence for the series $\sum_{n=1}^{\infty} \frac{4^n x^n}{7^{n+2}}$

$\left|\frac{4x}{7}\right| < 1$

$|4x| < 7$

$|x| < 7/4$

A) $7/4$

B) $\{0\}$

C) $4/7$

D) $(-\infty, \infty)$

E) Some other answer

c) The interval of convergence for the series $\sum_{n=0}^{\infty} \frac{(x+1)^n}{4^{n+1}}$

$|r| = \left|\frac{x+1}{4}\right| < 1$

A) $[-5,3]$

B) $(-5,3)$

C) $[-5,3)$

D) $(-5,3]$

Geometric series so we don't need to test the endpoints

$|x+1| < 4$

$-4 < x+1 < 4$

$-5 < x < 3$

d) The power series for $f(x)=\dfrac{1}{x+5}$ is:

$$\frac{1/5}{1-(-x/5)} = \sum_{n=0}^{\infty} \frac{1}{5}\left(\frac{-x}{5}\right)^n$$

$$\sum_{n=0}^{\infty} \frac{(-1)^n x^n}{5^{n+1}}$$

A) $\displaystyle\sum_{n=0}^{\infty} \frac{x^n}{5^{n+1}}$

B) $\displaystyle\sum_{n=0}^{\infty} \frac{(-1)^n 5^n}{x^{n+1}}$

C) $\displaystyle\sum_{n=0}^{\infty} \frac{(-1)^n x^n}{5^{n+1}}$

D) $\displaystyle\sum_{n=0}^{\infty} \frac{5^n}{x^{n+1}}$

E) Some other answer

e) The third degree polynomial for the series $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^n x^{2n+1}}{e^n\, n!}$ is

A) $x+\dfrac{x^3}{e}-\dfrac{x^5}{2e^2}$

B) $\dfrac{-x^3}{e}$

$n=1$ $\quad \dfrac{-x^3}{e(1!)}$ ← third degree

C) $x-\dfrac{x^3}{e}$

D) $\dfrac{-x^3}{e}+\dfrac{x^5}{2e^2}-\dfrac{x^7}{6e^3}$

E) Some other answer

f) The foci for the hyperbola $\dfrac{(x-1)^2}{7}-\dfrac{(y+2)^2}{9}=1$ are

Center $(1,-2)$

A) $(5,-2)$, $(-3,-2)$

B) $(1\pm\sqrt{2},-2)$

C) $(1,2)$, $(1,-6)$

D) $(1,-2\pm\sqrt{2})$

E) Some other answer

$c^2=a^2+b^2$

$c^2=7+9=16$

$c=4$

foci $(1\pm4,-2)$

g) The integral of the series $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$ is

A) $C + \sum_{n=0}^{\infty} \frac{(-1)^n (2n) x^{2n-1}}{(2n)!}$

B) $C + \sum_{n=0}^{\infty} \frac{(-1)^n (2n+1) x^{2n}}{(2n)!}$

C) $C + \sum_{n=1}^{\infty} \frac{(-1)^n (2n) x^{2n-1}}{(2n)!}$

D) $C + \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)(2n)!}$

E) Some other answer

h) The derivative of the series $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n)!}$ is

A) $\sum_{n=0}^{\infty} \frac{(-1)^n (2n) x^{2n}}{(2n)!}$

B) $\sum_{n=0}^{\infty} \frac{(-1)^n (2n+1) x^{2n}}{(2n)!}$

C) $\sum_{n=1}^{\infty} \frac{(-1)^n (2n+1) x^{2n-1}}{(2n)!}$

D) $\sum_{n=1}^{\infty} \frac{(-1)^n x^{2n+2}}{(2n+2)(2n)!}$

E) Some other answer

i) Which method can be used to express $f(x) = \frac{1}{4+x}$ centered at $a = 0$ as a series?

A) Binomial Series
B) MacLaurin Series
C) Geometric Series
D) Taylor Series
E) All of the given methods can be used.

j) Which of the following is not a conic section?

A) $\frac{x^2}{4}+\frac{y^2}{7}=1$

B) $3x^2=4y^2$ $3x^2-4y^2=0$

C) $x^2=y$

D) $x^2+y^2=1$

E) They are all conic sections.

2. (2 points each) True or False **Circle the correct answer**

a) **TRUE** **FALSE** The directrix for the parabola $x^2-4x-4y=$ is $y=-2$

b) **TRUE** **FALSE** When using the Ratio Test on a power series, the limit $L=0$ means that the Radius of convergence is 0. $L=0$ means $R=\infty$

c) **TRUE** **FALSE** The asymptotes for the hyperbola $\frac{(x-1)^2}{16}-\frac{(y+2)^2}{25}=1$ are $y=\pm\frac{5}{4}x$

d) **TRUE** **FALSE** The Maclaurin series representation for $f(x)=x\cos(x^3)$ is $\sum_{n=0}^{\infty}\frac{(-1)^n x^{6n+1}}{(2n)!}$.

e) **TRUE** **FALSE** The foci of the hyperbola $4x^2-4(y-1)=36$ are $(\pm 3\sqrt{2},1)$

not a hyperbola

f) **TRUE** **FALSE** The Taylor series for $f(x)=\sin x$ at $x=\pi/6$ is $\sum_{n=0}^{\infty}\frac{(-1)^n(x-\pi/6)^{2n+1}}{(2n+1)!}$.

g) **TRUE** **FALSE** $e = 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{24} \ldots$ $e = 1 + 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{24} \ldots$

h) **TRUE** **FALSE** A parabola can never intersect its diretrix.

i) **TRUE** **FALSE** To find the foci of a hyperbola, use the formula $c^2 + b^2 = a^2$

$a^2 + b^2 = c^2$

j) **TRUE** **FALSE** The radius of convergence for the series $\sum_{n=0}^{\infty} \frac{(-1)^n x^n}{4^n n!}$ is 4.

$R = \infty$

3. (1 point each) True of False **Circle the correct answer**

a) **TRUE** **FALSE** $9x^2 + 4x = y^2 + 6$ is an ellipse. its a hyperbola

$9x^2 + 4x - y^2 = 6$

b) **TRUE** **FALSE** $\frac{9!}{9} = 8!$

c) **TRUE** **FALSE** We can use the geometric series test to find the radius and interval of convergence for $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{5^{n+1}}$ $|r| = \left|\frac{x^2}{5}\right| < 1$

d) **TRUE** **FALSE** The endpoints of the Interval of Convergence for a series must always be tested to see if they are included or not. Not if a geometric series

e) **TRUE** **FALSE** $T_3(x)$ is the third partial sum for a Taylor Series.

MATH 152
Mrs. Bonny Tighe

EXAM IIIA

PART ONE
11.8-11.11, 10.5
50 points

NAME Solutions

TA_____ Wed.11/25/2019

1. (3 points each) **Circle the letter for the correct answer**

a) The series which is equivalent to $\sum_{n=1}^{\infty} \frac{(-1)^n x^{2n}}{5^n\, n!}$

A) $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{5^n\, n!}$

B) $\sum_{n=0}^{\infty} \frac{(-1)^{n+1} x^{2n+1}}{5^{n+1}\,(n+1)!}$

C) $\sum_{n=0}^{\infty} \frac{(-1)^{n-1} x^{2n-1}}{5^{n-1}\,(n-1)!}$

D) $\sum_{n=0}^{\infty} \frac{(-1)^{n-1} x^{2n+2}}{5^{n+1}\,(n+1)!}$

E) Some other answer

b) The sum of the series $\sum_{n=0}^{\infty} \frac{(-1)^n (\pi)^{2n}}{9^n\,(2n)!}$

$\sum_{n=0}^{\infty} \frac{(-1)^n (\pi/3)^{2n}}{(2n)!} = \cos \pi/3 = 1/2$

A) $\pi/3$

B) $3\sqrt{3}/2$

C) $\sqrt{3}/2$

D) $1/2$

E) Some other answer

c) Which of the following is not a conic section?

A) $\frac{x^2}{4} + \frac{y^2}{7} = 1$

B) $3x^2 = 4y^2$ $\quad 3x^2 - 4y^2 = 0$

C) $x^2 = y$

D) $x^2 + y^2 = 1$

E) They are all conic sections.

d) The derivative of the series $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n)!}$ is

A) $\sum_{n=0}^{\infty} \frac{(-1)^n (2n) x^{2n}}{(2n)!}$

B) $\sum_{n=0}^{\infty} \frac{(-1)^n (2n+1) x^{2n}}{(2n)!}$

C) $\sum_{n=1}^{\infty} \frac{(-1)^n (2n+1) x^{2n-1}}{(2n)!}$

D) $\sum_{n=1}^{\infty} \frac{(-1)^n x^{2n+2}}{(2n+2)(2n)!}$

E) Some other answer

e) The third degree polynomial for the series $\sum_{n=1}^{\infty} \frac{(-1)^n x^{2n+1}}{e^n n!}$ is

A) $x - \frac{x^3}{e} + \frac{x^5}{e^2(2)}$

B) $1 - \frac{x}{e} + \frac{x^2}{e^2} - \frac{x^3}{e^3(6)}$

C) $\frac{-x^3}{e}$ third degree

D) $1 - \frac{x^3}{e} + \frac{x^5}{2e^2}$

E) Some other answer

f) The foci for the hyperbola $\frac{(x-1)^2}{7} - \frac{(y+2)^2}{9} = 1$ are

A) $(5,-2)$, $(-3,-2)$

B) $(1 \pm \sqrt{2}, -2)$

C) $(1,2)$, $(1,-6)$

D) $(1, -2 \pm \sqrt{2})$

E) Some other answer

$c^2 = a^2 + b^2$

$c^2 = 7 + 9 = 16$

$c = 4$

$(1,-2)$ center

$(1 \pm 4, -2)$ foci

g) The integral of the series $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$ is

A) $C + \sum_{n=0}^{\infty} \frac{(-1)^n (2n) x^{2n-1}}{(2n)!}$

B) $C + \sum_{n=0}^{\infty} \frac{(-1)^n (2n+1) x^{2n}}{(2n)!}$

C) $C + \sum_{n=1}^{\infty} \frac{(-1)^n (2n) x^{2n-1}}{(2n)!}$

D) $C + \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)(2n)!}$

E) Some other answer

h) The radius of convergence for the series $\sum_{n=1}^{\infty} \frac{4^n x^n}{7^{n+2}}$

$\left|\frac{4x}{7}\right| < 1$

$|x| < 7/4$

A) $7/4$

B) $\{0\}$

C) $4/7$

D) $(-\infty, \infty)$

E) Some other answer

i) The interval of convergence for the series $\sum_{n=0}^{\infty} \frac{(x+1)^n}{4^{n+1}}$

A) $[-5,3]$

B) $(-5,3)$

C) $[-5,3)$

D) $(-5,3]$

geometric series

$r = \left|\frac{x+1}{4}\right| < 1$

don't need to test endpoints

$|x+1| < 4$

$-4 < x+1 < 4$

$-5 < x < 3$

j) Which method can be used to express $f(x) = \frac{1}{4+x}$ as a series?

A) Binomial Series
B) MacLaurin Series
C) Geometric Series
D) Taylor Series
E) All of the given methods can be used.

2. (2 points each) True or False **Circle the correct answer**

a) **TRUE** **FALSE** The Taylor series for $f(x) = \sin x$ at $x = \pi/6$ is
$\sum_{n=0}^{\infty} \frac{(-1)^n (x - \pi/6)^{2n+1}}{(2n+1)!}$

b) **TRUE** **FALSE** When using the Ratio Test on a power series, the limit $L = 0$ means that the Radius of convergence is 0.

if $L=0$ then $R=\infty$

c) **TRUE** **FALSE** The asymptotes for the hyperbola $\frac{(x-1)^2}{16} - \frac{(y+2)^2}{25} = 1$ are
$y = \pm \frac{5}{4} x$

$(y+2) = \pm 5/4 (x-1)$

d) **TRUE** **FALSE** The Maclaurin series representation for $f(x) = x\cos(x^3)$ is $\sum_{n=0}^{\infty} \frac{(-1)^n x^{6n+1}}{(2n)!}$.

e) **TRUE** **FALSE** The foci of the hyperbola $4x^2 - 4(y-1) = 36$ are
$(\pm 3\sqrt{2}, 1)$

not a hyperbola

f). **TRUE** **FALSE** The directrix for the parabola $x^2 - 4x - 4y =$ is $y = -2$

g) **TRUE** **FALSE** $e = 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{24} \ldots$ $= 1 + 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{24} \ldots$

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$R = \infty$

3. (1 point each) True of False **Circle the correct answer**

a) **TRUE** **FALSE** $9x^2 + 4x = 4y^2 + 6$ is an ellipse. it is a hyperbola

b) **TRUE** **FALSE** $\frac{11!}{11} = 10!$

c) **TRUE** **FALSE** We can use the geometric series test to find the radius and interval of convergence for $\sum_{n=0}^{\infty} \frac{(-1)^n x^{3n}}{5^{n+1}}$ $|r| = \left|\frac{x^3}{5}\right| < 1$

d) **TRUE** **FALSE** The endpoints of the Interval of Convergence for a series must always be tested to see if they are included or not. not of a geometric series

e) **TRUE** **FALSE** $T_3(x)$ is the third partial sum for a Taylor Series.